Comparison of AlexNet and DHVT on the Same Image Classification Dataset

Course Presentation

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Outline

1. Introduction
2. Dataset and Workflow
3. Models
4. Training and Difficulties
5. Results
6. Conclusion

Project Goal

- The project compares a classical convolutional model and a transformer-style model under the same dataset setting.
- The two models are trained, saved, reloaded, and evaluated in the same notebook workflow.
- The main objective is to analyze not only accuracy, but also training behavior, confidence, and difficult examples.

Why Compare AlexNet and DHVT?

Model	Role in the study
AlexNet	CNN baseline with strong image inductive bias and a clear layer-by-layer structure.
DHVT	Transformer-style model adapted to small datasets through overlapped patch embedding and DAFF blocks.

- This comparison contrasts local convolutional processing with token-based attention.
- Both models use the same dataset configuration to keep the experiment fair.

Dataset Options

Option	Description
`CIFAR-10`	Standard 10-class benchmark used as the main baseline dataset.
`CIFAR-100`	Same image format with more classes and higher difficulty.
`custom_folder`	User dataset with `train/class_name` and `val/class_name` folders.

- Both models can be trained on the same selected dataset.
- Custom inference images are stored separately in `test_images/`.

Shared Experimental Workflow

Step	Procedure
1	Choose local runtime or Colab runtime.
2	Select the dataset and preprocessing settings.
3	Train AlexNet and DHVT on the same split.
4	Save the best checkpoints for both models.
5	Reload the saved models for evaluation and inference.
6	Compare training curves and qualitative outputs.

AlexNet

- AlexNet uses stacked convolutions, ReLU, pooling, and fully connected layers.
- In the notebook, it serves as the CNN baseline for the comparison.
- The convolution operation is written as:

$$Y_{i,j} = \sum_u \sum_v K_{u,v} X_{i+u,j+v}$$

$$\mathrm{ReLU}(x) = \max(0, x)$$

- This structure emphasizes local feature extraction and spatial hierarchy.

DHVT

- DHVT is a transformer-style architecture adapted to smaller datasets.
- The model includes SOPE, class token, positional embeddings, attention blocks, and DAFF.
- Patch-token embedding can be summarized as:

$$X \in \mathbb{R}^{B \times C \times H \times W} \rightarrow E \in \mathbb{R}^{B \times N \times d}$$

$$Z_0 = [x_{\text{cls}}; E], \qquad Z = Z_0 + P$$

- The model is trained from scratch rather than used only for fine-tuning.

Attention and DHVT Block

$$Q = ZW_Q, \qquad K = ZW_K, \qquad V = ZW_V$$

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

$$Z' = Z + \mathrm{MHSA}(\mathrm{LN}(Z))$$

$$Z'' = Z' + \mathrm{DAFF}(\mathrm{LN}(Z'))$$

- The DAFF block adds a convolution-aware feed-forward stage.
- This design helps the transformer preserve useful locality for small datasets.

Training Strategy

- Both models are trained under the same dataset configuration.
- Checkpoints are saved and reloaded inside the notebook.
- The objective function is the cross-entropy loss:

$$\mathcal{L} = -\sum_{c=1}^{C} y_c \log(\hat{p}_c)$$

- Validation accuracy is monitored to keep the best weights for each model.

Main Difficulties and Solutions

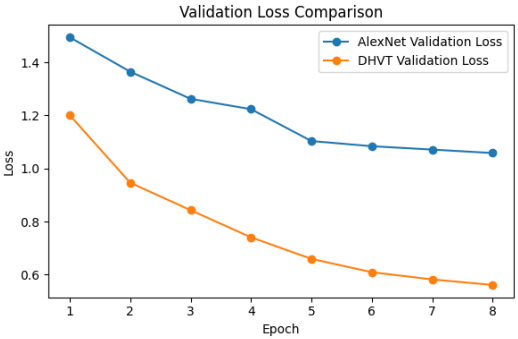
Difficulty	Solution
Fair comparison	Use one notebook and one shared dataset configuration.
Small-dataset transformer training	Use a DHVT-style architecture trained from scratch.
Homework-style presentation	Keep the workflow step-by-step in the notebook.
Reusability	Support multiple datasets and custom folders.
Long training runs	Print intermediate progress during training.

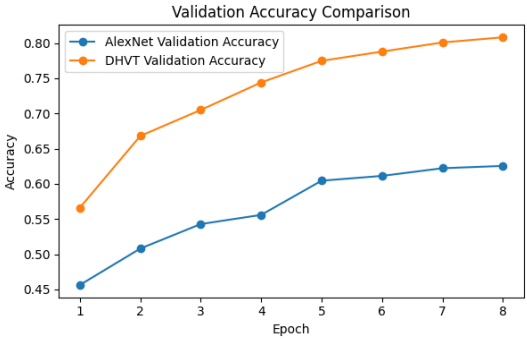
Main Numerical Results

Metric	AlexNet	DHVT
Best validation accuracy	0.6254	0.8081
Number of parameters	58,322,314	6,504,202
Mean confidence	0.6150	0.8247
Mean confidence on correct predictions	0.7000	0.8285
Mean confidence on incorrect predictions	0.4712	0.8289

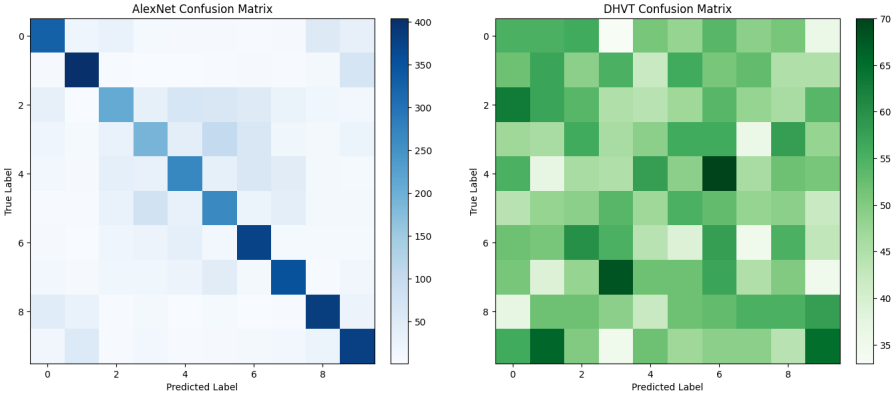
- DHVT achieved higher validation accuracy with far fewer parameters.
- This is the strongest quantitative result currently available.

Validation Curves





Confusion Matrices



- This figure illustrates the intended class-level comparison.
- It should be regenerated after rerunning the corrected evaluation cells.

Misclassified Examples

True: ship
AlexNet: automobile
DHVT: ship

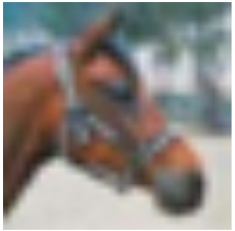
True: frog
AlexNet: deer
DHVT: frog

True: automobile
AlexNet: truck
DHVT: automobile

True: cat
AlexNet: dog
DHVT: cat

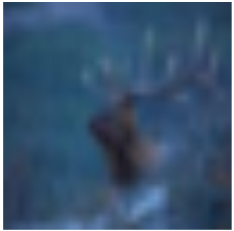
True: dog
AlexNet: frog
DHVT: dog

True: horse
AlexNet: truck
DHVT: horse

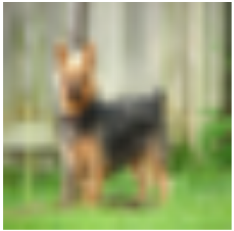


Both Models Wrong

True: deer
AlexNet: bird
DHVT: bird

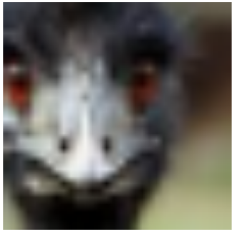


True: dog
AlexNet: deer
DHVT: deer



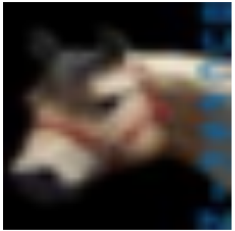
True: dog
AlexNet: cat
DHVT: cat

True: bird
AlexNet: cat
DHVT: cat

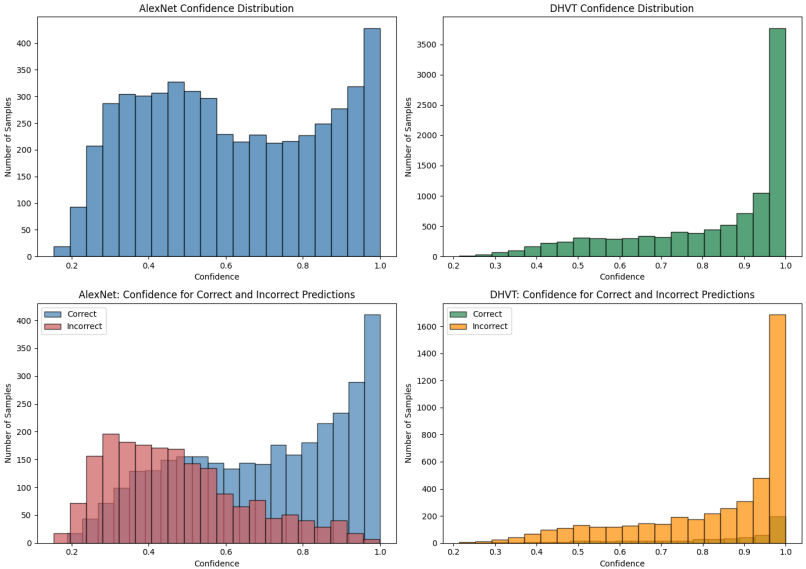


True: airplane
AlexNet: frog
DHVT: frog

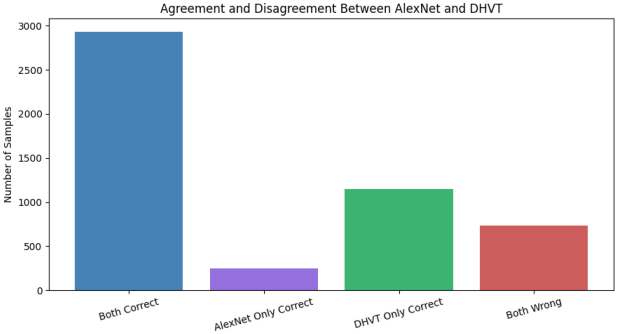
True: horse
AlexNet: truck
DHVT: cat



Confidence Distribution



Agreement Analysis



Case	Ratio
Both correct	0.5803
AlexNet only	0.0483
DHVT only	0.2265
Both wrong	0.1450

Evaluation Note

- A label-alignment bug was identified in some post-training analysis cells.
- The issue came from zipping two evaluation loaders that used different batch sizes.
- Training curves and parameter counts remain valid.
- Post-training visual analyses should be rerun in Colab with the corrected notebook cells.

Conclusion

- DHVT is the stronger model in the current experiment.
- It achieves better validation accuracy while using fewer parameters than AlexNet.
- The notebook now supports training, saving, reloading, inference, and visual comparison in one workflow.
- The project also shows that model evaluation should include curves, confidence behavior, and qualitative error analysis.

Future Work

- Rerun the corrected post-training analysis cells and refresh the exported figures.
- Extend the comparison to another dataset such as CIFAR-100 or a custom image folder.
- Add transformer-specific visualizations such as attention maps after the evaluation pipeline is stable.

References

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An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale.
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